



AWS Data Engineering Crash Course



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Introduction to AWS Data Engineering

Data Warehouse

Building a structured data repository for analytical processing using Amazon Redshift.

Data Lake

Creating a flexible storage solution for diverse data types using Amazon S3.



 Data Lake

 Data Warehouse

A storage repository for raw data, utilizing Amazon S3 for semi-structured data.

A system for reporting and data analysis, using Amazon Redshift for structured data.





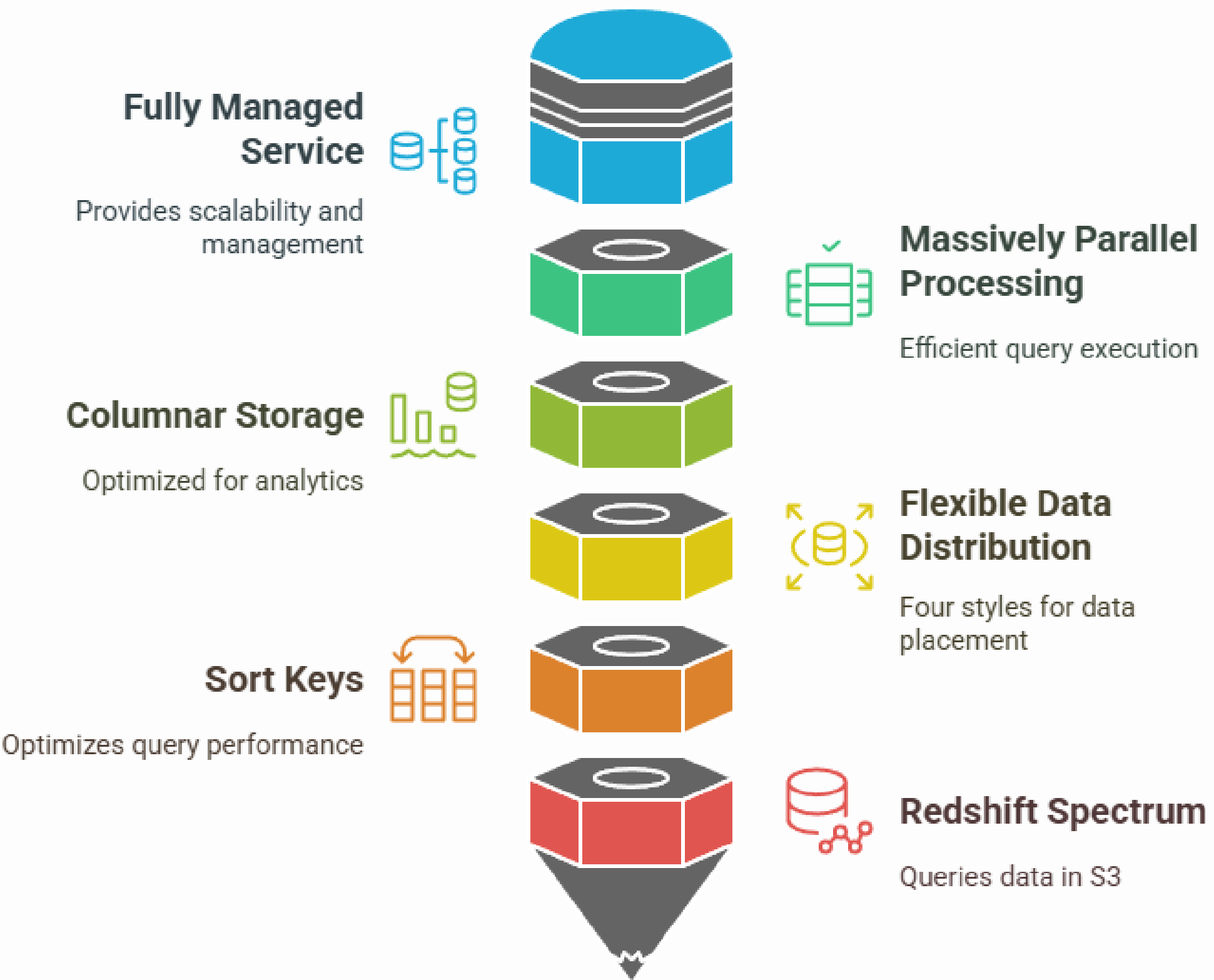
Chapter 1: Amazon Redshift (Overview)

Amazon Redshift Features Overview



amazon
REDSHIFT

Fundamentals



Chapter 2: AWS Glue (Overview)



Fundamentals

Serverless AWS Service

Don't worry about servers/nodes procurement or maintenance



Data Catalog

Metadata Repository used for Glue , Athena, Lake formation.
Can also be used by Spark, Hive on EMR.
Or external databases on Redshift



Spark ETL

Primarily used for running ETL jobs leveraging Spark



Crawlers

Discover underlying data automatically from the source.



Python Jobs

Also used for building & running python jobs.



Glue Studio

Visual interface for creating ETL jobs





Apache Hudi



Open Source Data Lake

Used for building data lakes



ACID Transactions

Support
INSERT/UPDATE/DELETE on
Hudi Tables



Versioning

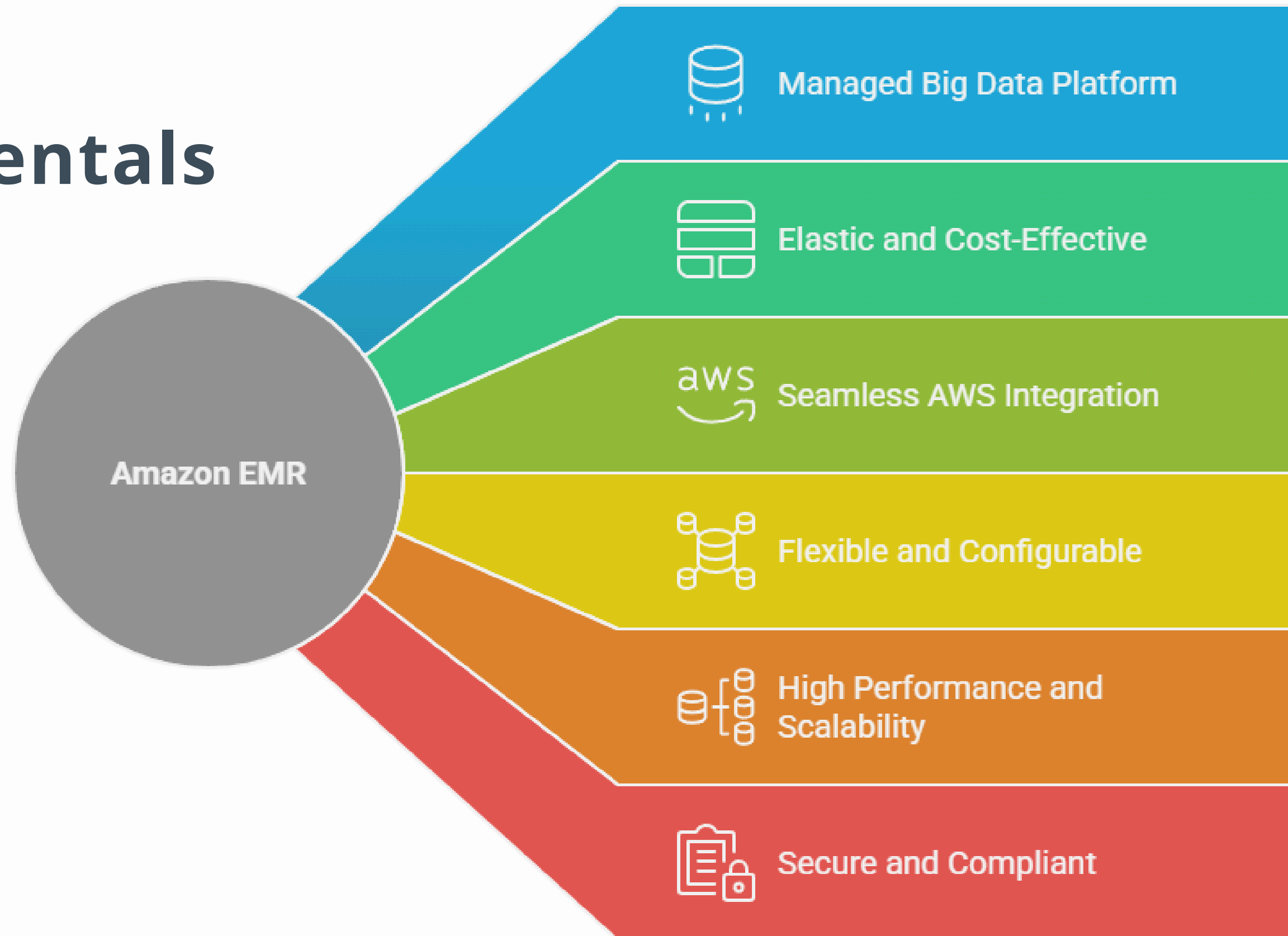
Supports versioning & rollback



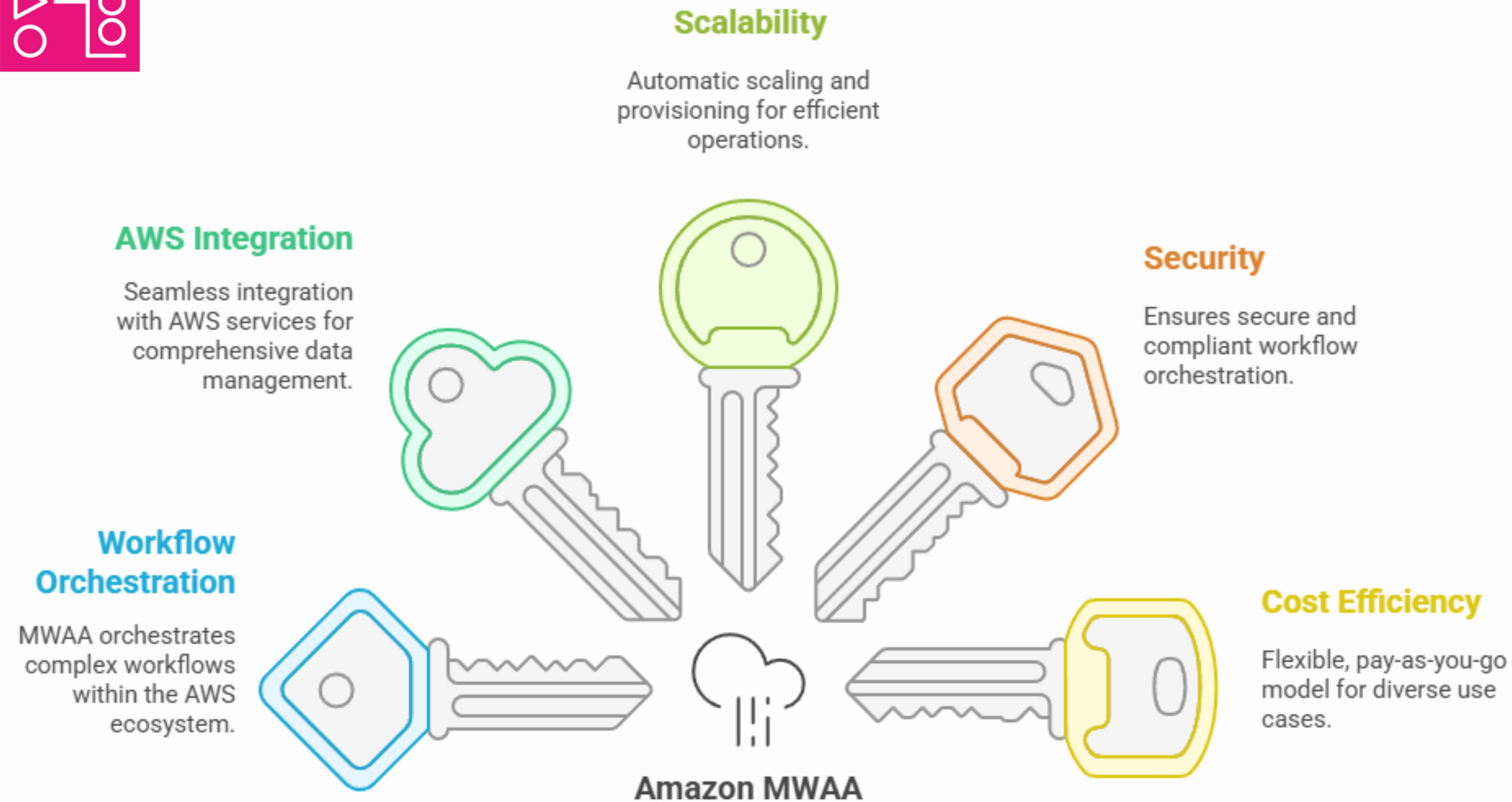
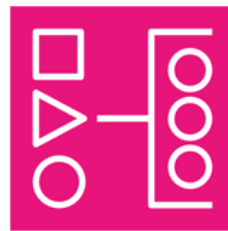
Readable by aws services

Seamless integration with
AWS Glue , Athena, Redshift

Fundamentals



Chapter 5: Managed Airflow (MWAA Overview)



Amazon Managed Workflows for Apache Airflow (MWAA)



Exercise 1: *AWS Glue* (hands-on)





Exercise 2: Data Lake - Glue + Hudi

(hands-on)





Exercise 3: Amazon EMR (hands-on)





Exercise 4: Amazon Redshift (hands-on)



Exercise 5: MWAA - run end to end pipeline (hands-on)